The Lighthouse Lantern Mystery

Whodunnit? · Right Triangles & Trig

Name: _____

On a foggy night at Gull Point Lighthouse, the great light went dark. The keeper climbed the stairs and gasped: the **Great Lens**, the giant glass that guides every ship to harbor, was gone!

Six people were near the lighthouse, each in a different spot, and there are six ways a heavy lens could be carried away. You are the harbor detective: find **who** took it, **where** they were, and **how** they got it out.

How to crack the case

1. Your teacher will post **10 clue cards** around the room. Each card has a math problem and four answer choices. Every choice is paired with a clue.
2. Solve the problem on the **Show Your Work** page. Find your answer among the choices and read the clue next to it. If your answer is not there, check your work.
3. Use the clue to cross out suspects, places or methods on the case board below. Write the card number beside what it cleared.
4. After all 10 cards, exactly one suspect, one place and one method are left. Those three boxes name the culprit. If you end up with none, or more than one, a clue went wrong: recheck your math!

The Suspects	The Places	The Methods
☐ Keeper Odell Lighthouse Keeper	☐ Lamp Room	☐ Rowed away in a dinghy
☐ Captain Marlow Ferry Captain	☐ Spiral Stairs	☐ Hidden in a fish crate
☐ Ms. Brine Fish Seller	☐ Harbor Dock	☐ Lowered by rope
☐ Dr. Gale Weather Scientist	☐ Fish Market	☐ Wrapped in a fishing net
☐ Nico Reyes Boat Builder	☐ Cliff Path	☐ Carried in a backpack
☐ Mr. Tern Birdwatcher	☐ Boathouse	☐ Floated in a barrel

The culprit is _____ **in** _____ **using** _____

Show Your Work

The Lighthouse Lantern Mystery · Show every step in the matching box.

Name: _____

1. Choice _____	**2.** Choice _____
3. Choice _____	**4.** Choice _____
5. Choice _____	**6.** Choice _____
7. Choice _____	**8.** Choice _____
9. Choice _____	**10.** Choice _____

Clue #1

The Lighthouse Lantern Mystery

A right triangle has legs 6 and 8. Find the hypotenuse.

A. $2\sqrt{7}$	Every backpack at the ferry gate was unzipped and searched. Just lunches and rain jackets.
B. 100	Keeper Odell was sketching the sea view from the Cliff Path. A hiker snapped his photo.
C. 14	Dr. Gale was checking the wind gauge in the Lamp Room. The weather log is stamped with the hour.
D. 10	Keeper Odell was polishing the brass in the Lamp Room. The visitor logbook shows his time.

Solve the problem, find your answer, and read its clue. Cross it off on your case board.

✂ cut here

Clue #2

The Lighthouse Lantern Mystery

A right triangle has a leg of 5 and a hypotenuse of 13. Find the other leg.

A. 8	The backpacks on the hook rack were all checked. Only sweaters and sandwiches inside.
B. $\sqrt{194}$	Captain Marlow was tying his boots on the Cliff Path. A jogger waved to him.
C. 18	Dr. Gale was climbing the Spiral Stairs with a barometer. Three tourists counted her steps.
D. 12	The dinghy is still tied up with the tide line untouched. No one rowed.

Solve the problem, find your answer, and read its clue. Cross it off on your case board.

Clue #3

The Lighthouse Lantern Mystery

An isosceles right triangle has legs of length 5. Find the hypotenuse.

A. $5\sqrt{2}$	Captain Marlow was loading passengers at the Harbor Dock. All twenty remember him.
B. $5\sqrt{3}$	Ms. Brine was selling mackerel from a cart on the Cliff Path. Her receipts have the time.
C. 10	Dr. Gale was talking to the crew at the Harbor Dock about the storm. The dockmaster wrote it down.
D. 25	The only backpack found was empty and flat. Nothing was carried in it.

Solve the problem, find your answer, and read its clue. Cross it off on your case board.

✂ cut here

Clue #4

The Lighthouse Lantern Mystery

In a 45°-45°-90° triangle the hypotenuse is 8. Find the length of a leg.

A. $8\sqrt{2}$	Dr. Gale was buying a crab roll at the Fish Market. The vendor remembers her umbrella.
B. $4\sqrt{2}$	Every fish crate was opened at the market. Only fish.
C. $4\sqrt{3}$	A backpack was weighed on the dock scale. It matched its owner's gear, nothing extra.
D. 4	Mr. Tern was watching puffins from the Cliff Path with binoculars. His bird list has the times.

Solve the problem, find your answer, and read its clue. Cross it off on your case board.

Clue #5

The Lighthouse Lantern Mystery

In a 30°-60°-90° triangle the shortest side is 4. Find the longer leg.

A. $4\sqrt{3}$	Ms. Brine was selling cod at the Fish Market. The cash register shows each sale.
B. 8	Dr. Gale was borrowing a tarp in the Boathouse. Nico Reyes signed her out.
C. $2\sqrt{3}$	Search dogs sniffed every backpack at the gate. None of them reacted.
D. $4\sqrt{2}$	Nico Reyes was measuring the Cliff Path fence for a new railing. His tape notes are dated.

Solve the problem, find your answer, and read its clue. Cross it off on your case board.

✂ cut here

Clue #6

The Lighthouse Lantern Mystery

In right triangle ABC with the right angle at C, AB = 5, BC = 3 and AC = 4. Find sin A.

A. $\frac{3}{4}$	Mr. Tern was sketching a gull nest on the Cliff Path until sunset. His sketchbook has the date.
B. $\frac{3}{5}$	The rope is coiled and dry. Nothing was lowered.
C. $\frac{4}{5}$	Dr. Gale was cleaning the lens in the Lamp Room with Keeper Odell. The brass is still shiny.
D. $\frac{5}{3}$	The straps on every backpack were clean and unworn. None had carried anything heavy.

Solve the problem, find your answer, and read its clue. Cross it off on your case board.

Clue #7

The Lighthouse Lantern Mystery

In right triangle ABC with the right angle at C, AB = 5, BC = 3 and AC = 4. Find tan A.

A. $\frac{4}{3}$	Dr. Gale was sending a radio report from the Harbor Dock. The signal log shows the time.
B. $\frac{4}{5}$	Nobody left the lighthouse with a backpack that day. The gate guard kept a tally.
C. $\frac{3}{4}$	Nico Reyes was sanding a hull in the Boathouse with two apprentices.
D. $\frac{3}{5}$	Captain Marlow was waiting out a gust on the Cliff Path. A kite flyer saw him there.

Solve the problem, find your answer, and read its clue. Cross it off on your case board.

✂ cut here

Clue #8

The Lighthouse Lantern Mystery

A 10-ft ladder leans against a wall and makes a 60° angle with the ground. How high up the wall does it reach?

A. $10\sqrt{3}$ ft	Keeper Odell was trimming weeds along the Cliff Path. The gardener's sheet backs him up.
B. $5\sqrt{2}$ ft	A backpack was opened at the lookout. It held only a thermos and a map.
C. $5\sqrt{3}$ ft	The fishing net is hanging, dry, and clean. Nothing was wrapped in it.
D. 5 ft	Dr. Gale was resting on the Spiral Stairs with a cup of tea. A lighthouse guide sat with her.

Solve the problem, find your answer, and read its clue. Cross it off on your case board.

Clue #9

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If $\sin \theta = \frac{1}{2}$ for an acute angle θ, find θ.

A. 60°	Dr. Gale was mending a rain gauge in the Boathouse. The boat builder's apprentices watched her.
B. 26.6°	The lost-and-found backpack was checked twice. It held only socks and a flashlight.
C. 45°	Ms. Brine was setting up a stall near the Cliff Path at noon. Her permit slip is stamped.
D. 30°	Mr. Tern was counting seagulls from the Spiral Stairs. His notebook has the times.

Solve the problem, find your answer, and read its clue. Cross it off on your case board.

✂ cut here

Clue #10

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Find x to the nearest tenth:

$$\tan 35° = \frac{x}{10}$$

A. 8.2	The backpack on the bench belonged to a tourist. It held only snacks and sunscreen.
B. 14.3	Dr. Gale was sampling chowder at the Fish Market. The stall owner kept her receipt.
C. 7.0	The barrels by the dock are sealed and heavy, so nothing floated away in them.
D. 5.7	Nico Reyes was carving a sign for the Cliff Path trailhead. Two hikers stopped to watch.

Solve the problem, find your answer, and read its clue. Cross it off on your case board.